Artifical Intelligence Basics Course Summary

Generated by Scribe.AI

December 9, 2024

1 Key Terms

1.1 Search Algorithms

- **Minimax Search**: A search algorithm for two-player zero-sum games that aims to find the best move for the current player, assuming the opponent also plays optimally.
- **Alpha-Beta Pruning**: A technique to reduce the search space in Minimax search by avoiding the evaluation of branches that cannot affect the optimal decision.

1.2 Machine Learning Paradigms

- **Perceptron**: A fundamental unit in neural networks, consisting of weighted inputs, a summation node, and an activation function that produces an output.
- **Reinforcement Learning**: A type of machine learning where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties.

1.3 AI Foundations

- **Turing Test**: An operational test for intelligent behavior, proposed by Alan Turing, where a human evaluator must distinguish between a human and a machine based on their textual responses.

1.4 Game AI

- **Game Tree**: A tree-like structure representing all possible sequences of moves in a game, with leaf nodes representing the outcomes.

2 Problem Types

2.1 Foundational AI Concepts

- **Understanding the Nature of Intelligence**: This problem explores the definition and characteristics of intelligence, both human and artificial. Different perspectives and definitions of AI are examined.
- **Evaluating Intelligent Behavior**: This problem focuses on developing methods for evaluating whether a machine exhibits intelligent behavior. The Turing Test and its limitations are discussed, along with counterarguments like Searle's Chinese Room argument.
- **Understanding the Components of AI Systems**: This problem focuses on identifying the key components of AI algorithms, including knowledge representation, decision-making processes, and the ability to generalize to new situations.

2.2 AI Challenges and Limitations

- **Overcoming the Limits of Search**: This problem addresses the computational challenges of exhaustive search in complex games like chess and Go. The exponential growth of search space is highlighted, leading to the need for machine learning techniques to supplement search.
- **Addressing AI Safety and Ethical Concerns**: This problem explores the potential risks and ethical implications of AI systems, including bias in algorithms and the vulnerability of neural networks to adversarial attacks.
- **Bridging the Gap Between Different AI Domains**: This problem investigates the commonalities and differences between seemingly disparate AI tasks, such as game playing and speech synthesis. The shared algorithmic components and challenges are analyzed.

2.3 AI Applications and Successes

- **Explaining the Success of AI in Game Playing**: This problem examines the factors contributing to the success of AI in game playing, including the use of deep learning, search algorithms, and the combination of both. Specific examples of successful AI game-playing systems are analyzed.

3 Algorithm Solutions

3.1 Game Playing AI

- **Minimax Search**: A recursive algorithm for two-player zero-sum games that explores the game tree to find the best move for the current player, assuming the opponent also plays optimally. It alternates between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha-Beta Pruning**: An optimization technique for Minimax search that reduces the search space by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha and beta values to prune branches.